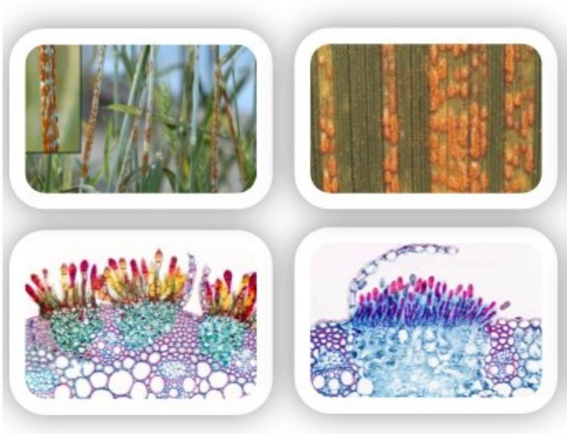
Technical Booklet No.02 (2020-21)

Rust of Wheat



सत्यमेव जयते

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Ministry of Agriculture & Farmer's Welfare
Department of Agriculture Cooperation & Farmer's Welfare
Integrated Pest Management Division
Directorate of Plant Protection Quarantine & Storage
NH-IV, Faridabad

Wheat Rust:

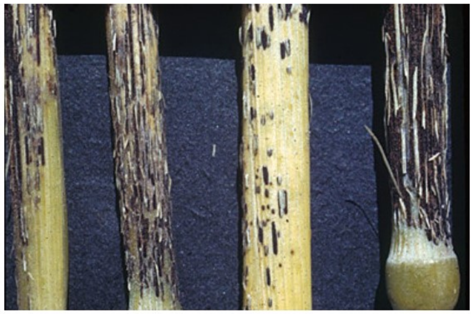
Wheat is major staple crop in India after Paddy. The total area under this crop is about 30 million ha. (Approx.) with an annual production of 103 million MT (Approx.). Though many biotic stresses disrupt wheat cultivation, wheat rust is the devastating one not only in India but also throughout the wheat growing countries of the world. Both Stem rust and stripe rust can cause up to 100% crop loss, whereas leaf rust can cause 45-50% crop loss, under favourable climatic conditions. In India stripe rust of wheat (caused by *Puccinia striiformis* f. sp. *tritici* is a threat in 10 million hectares of Northern India, whereas stem rust (caused by *P. graminis* f. sp. *tritici* threatens about 7 million hectares of Central and Peninsular India. In contrast, leaf rust (caused by *P. triticina*) is prevalent wherever wheat is grown.

Black or stem rust

It is caused by *P. graminis* f. sp. *tritici*. At one time, it was a feared disease of wheat crop worldwide. The fear of stem rust was understandable because an apparently healthy crop three weeks before harvest could be reduced to a black tangle of broken stems and shrivelled grain by harvest.

Symptoms

Stem Rust symptom on stem



Black teleutospore on stem rust

- After 7 to 15 days of infection, symptoms are produced on almost all aerial parts of the wheat plant but are most common on stem, leaf sheaths and leaf surfaces.
- Uredial pustules (or sori) are oval to spindle shaped and dark reddish brown (rust) in colour.
- They erupt through the epidermis of the host and are surrounded by tattered host tissue.
- The pustules are dusty in appearance due to vast number of spores produced. Spores are readily released, when touched.
- As the infection advances, teliospores are produced in the same pustule. Colour of the pustule changes from rust to black as teliospore production progresses.

- If more pustules are produced, stem become weakened and lodge.
- The pathogen attacks other host (barberry) to complete its life cycle. Symptoms are very different on this woody host.
- Other spores are Pycnia (spermagonia) produced on the upper leaf surface of barberry which appear as raised orange spots. Small amounts of honeydew that attracts insects are produced in this structure.
- Aecia, produced on the lower leaf surface, are yellow. They are bell-shaped and extend as far as 5 mm from the leaf surface.

Epidemiology:

The minimum, optimum and maximum temperatures for spore germination are 2°, 15° to 24°, and 30°C, respectively and for sporulation, 5°, 30° and 40°C, respectively. Stem rust is more important on late-sown and maturing wheat cultivars, and at lower altitudes. Spring-sown wheat is particularly vulnerable in the higher latitudes if sources of inoculum are located downwind. In warm humid climates, stem rust can be especially severe due to the long period of favourable conditions for disease development when a local inoculum source is available. Maximum infection is obtained with 8 to 12 hours of dew at 18°C followed by 10 000+ lux of light while the dew slowly dries and the temperature rises to 30°C.

Alternate hosts

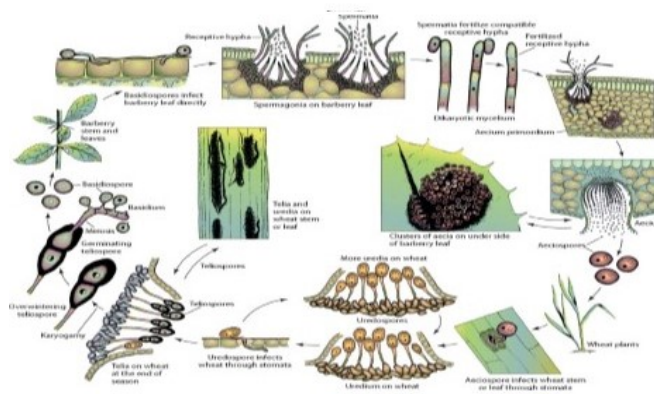
The main alternate host for *P. graminis* is *B. vulgaris*. However, barberries species found in India do not play any role in the perpetuation of stem rust in India

Life cycle

The life cycle of *P. graminis* f. sp. *tritici* consists of continual uredinial generations. The fungus spreads by airborne urediniospores from one wheat plant to another and from field to field. Primary inoculum may originate locally (endemic) from volunteer plants or be carried long distances (exodemic) by wind and deposited by rain. Urediniospore germination starts in one to three hours at optimum temperatures in the presence of free water. The moisture or dew period must last six to eight hours at favourable temperatures for the spores to germinate. As the host matures, telia are produced directly from urediniospore infections or teliospores can be produced in a mature

uredinial pustule.The teliospore germinates, undergoes meiosis and produces a four-celled basidium. The hyaline basidiospore is windborne short distances (metres) to the barberry bush.

Urediospores on leaf (Stem Rust)



Disease cycle of *P. graminnis* f. sp. tritici

Basidiospores germinate and penetrate directly the barberry plant. Infection by a basidiospore results in the production of a pycnium. Aeciospores are hydroscopically released from the aecia and are airborne to wheat over distances of metres to perhaps a few kilometres. Under field conditions, where temperatures vary greatly, the cycle can be either lengthened or shortened. Generally, lower temperatures in the field, at least in the early stages of the crop cycle, tend to lengthen the latent period.

Brown or leaf rust

Of the rust diseases of wheat, the most common these days is called leaf or brown rust and is caused by *P. triticina* . Leaf rust occurs to some extent wherever wheat is grown. Losses due to leaf rust are usually small (less than 10 percent), but can be severe (30 percent or more) under favourable environmental conditions.

Symptoms

♦ The most common site for symptoms is on leaf blades, however, sheaths, glumes and awns may occasionally become infected and exhibit symptoms.

♦ Uredia are seen as small, circular orange blisters or pustules on the upper surface of leaves.

symptom of Brown Rust on leaf

Foliar symptom of Brown Rust

♦ Orange spores are easily dislodged and may cover clothing, hands or implements.

♦ When the infection is severe, leaves dry out and die.

♦ Since inoculum is blown into a given area, symptoms are often seen on upper leaves first.

♦ As plants mature, the orange urediosporesare are replaced by black teliospores. Pustules containing these spores are black and shiny since the epidermis does not rupture.

Epidemiology

The fungus can infect with dew periods of three hours or less at temperatures of about 20°C; however, more infections occur with longer dew periods. Most of the severe epidemics occur when uredinia and/or latent infections survive the winter at some threshold level on the wheat crop, or where spring-sown wheat is the recipient of exogenous inoculum at an early date, usually before heading. Severe epidemics and losses can occur when the flag leaf is infected before anthesis. However, the latent period (uredinial) is approximately three to four days longer, and teliospore production starts shortly after initial urediniospore production.

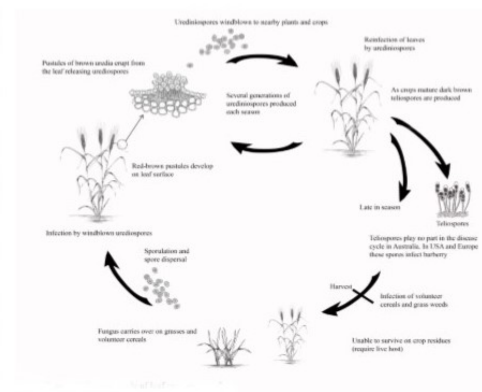
Alternate hosts

The primary alternate host of *P. triticina*, including the durum attacking populations, is *T.speciosissimum*. Whereas *A. agregata*, *A.undulata*, *Echiumglomeratum* and *Lycopsisarvensis* (Boraginaceae) are the alternate hosts for the leaf rust on wild wheat (*Triticum* [*Aegilops*] spp.) and rye.

Life cycle

The alternate host currently provides little direct inoculum of *P. triticina* to wheat. The pathogen survives the period

between wheat crops in many areas on a green-bridge of volunteer (self-sown) wheat. Inoculum in the form of uredinio spores can be blown by winds from one region to another. Teliospores can germinate shortly after development, and basidiospore infection can occur throughout the wheat-growing cycle. Urediniospores initiate germination 30 minutes after contact with free water at temperatures of 15° to 25°C. Spore germination to sporulation can occur within a seven- to ten-day period at optimum and constant temperatures. At low temperatures (10° to15°C) or diurnal fluctuations, longer periods are necessary. Maximum sporulation is reached about four days following initial sporulation (at about 20°C).



The teliospores of *P. triticina* are formed under the epidermis with unfavourable conditions or senescence and remain with the leaves. Leaf tissues can be dispersed or moved by wind, animals or humans to considerable distances. Basidiospores are formed and released under humid conditions, which limit their spread. Aeciospores are more similar to urediniospores in their ability to be transported by wind currents, but long-distance transport has not been noted for some reason.

Yellow or stripe rust

Stripe or yellow rust of wheat caused by *P. striiformis* f. sp. *tritici* can be as damaging as stem rust. However, stripe rust has a lower optimum temperature for development that limits it as a major disease in many areas of the world. Stripe rust is principally an important disease of wheat during the winter or early spring or at high elevations

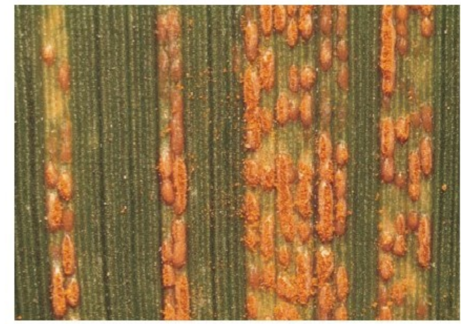
Symptom

♦ Initially, symptoms are yellowish flecks on leaves.

♦ On susceptible varieties, pustules containing yellow-orange spores erupt from leaves.

Pustules are clustered on seedling leaves, while pustules on mature leaves occur in a linear, stripe-like pattern.

Stripe like symptom on leaf



Pustule on leaf

♦ Later in the season, yellow-orange fungal spores turn black and remain attached to leaf tissue.

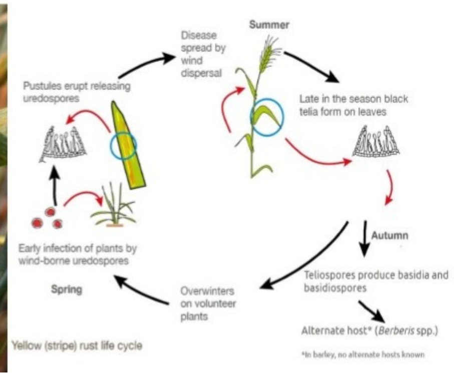
♦ Symptoms can be present from seedling stages through ripening. Stripe rust pustules form a noticeable striped pattern on mature leaves and are more yellow than stem rust spores.

♦ The teliospores are also arranged in long stripes and are dull black in colour.

Epidemiology

Puccinia striiformis has the lowest temperature requirements of the three wheat rust pathogens. Minimum, optimum and maximum temperatures for stripe rust infection are 0°, 11° and 23°C, respectively. *Puccinia striiformis* frequently can actively overwinter on autumn-sown wheat. Primary infections are caused by wind-borne urediospores that may have travelled long distances. The disease may develop rapidly when free moisture (rain or dew) occurs and temperatures range between 10-20°C.At temperatures above 25°C,the production of urediospores is reduced or ceases and black teliospores are often produced.

Life cycle



Puccinia striiformis is most likely a hemiform rust in that the life cycle seems only to consist of the uredinial and telial stages. Uredia develop in narrow, yellow, linear stripes mainly on leaves and spikelets. When the heads are infected, the pustules appear on the inner surfaces of glumes and lemmas. The urediniospores are yellow to orange in colour, more or less spherical, echinulate and 28 to 34 μm in diameter. Teliospores are dark brown, two-celled and similar in size and shape to those of *P. triticina*. Stripe rust populations can exist, change in virulence and result in epidemics independent of an alternate host. Urediniospores are the only known source of inoculum for wheat, and they germinate and infect at cooler temperatures.

Simple Microscopy of Wheat Rust:

Identification of plant pathogens/disease largely depends on symptoms appearance on the plants. However, sometimes proper symptoms are not always appearing on plant surface due to different factors but the pathogens may always present on the plant. Simple microscopy always helps to determine the different fungal pathogens by observing reproductive structures presents on the plants. As rust pathogens are obligate in nature, thus it cannot be cultured in artificial medium. Microscopy of the rust associated sample will help to understand the disease stages present in the field and also helps to determine the future course of action also.

Sample Collection:

Sample collection is an important task of microscopy. During survey samples may be collected as cautiously as possible. Plant parts with symptomatic appearance may be collected. Newly appeared symptom along with old symptom bearing samples must be collected and preserved in such a way that the sample must not be damaged during transportation. Samples of leaf/stem/grains must not be mixed with soiled root. Sample may be kept in a air tight PP Bag filled with air/ container to better preservation. Fresh sample always give a good result in simple/ light microscopy.

Preparation:

1. Sample need to cut 1.5-2.0 cm in length.
2. Surface sterilization with 1% sodium hypochlorite solution

3. Wash with distilled water.
4. Carrot/ potato blocks may be prepared to transverse sectioning of the samples.
5. Put the sample in carrot/potato block and slice it with a razor/fine blade. Finer the slicing of sample greater the chance of seeing pathogenic structures in intact.
6. Put the finely cut samples in watch glass and remove carrot/potato slice by a fine tip brush.
7. Stain the sample with simple lactophenol-cotton blue solution for 1-2 min
8. Wash it in lacto-phenol solution or in simple glycerine
9. Mount the sample pieces on a slide using a drop of lactophenol/ glycerine and cover it with a cover slip.
10. Observe it under Microscope.

Differential staining methods can also be followed using Safranin stain. In this case plant tissue will also be stained and fungi colour will be different.

Management of Rust Disease:

Cultural control:

- Follow mixed cropping and crop rotation
- The most effective, and the only practical, means of control of wheat stem rust is through the use of wheat varieties resistant to infection by the pathogen
- Eradication of barberry has reduced losses from stem rust by eliminating the early season infections on wheat in areas where uredospores cannot overwinter, and by reducing the opportunity for the development of new races of the stem rust fungus through genetic recombination on barberry
- Damage by the stem rust fungus is usually lower in fields in which heavy fertilization with nitrate forms of nitrogen and dense seeding have been avoided

Chemical Management:

Pesticides recommended for management of different rust by CIBRC are as follows-

Fungicide/ Fungicide combination name	Common Name of the disease	Dosage per ha			Waiting period from last application to harvest (in days)
		a.i. (g)	Formulation (g/ml)/ %	Dilution in water (L)	
Kresoxim-methyl 44.3% SC	Rust	250	500	500	25
Mancozeb 75% WP	Brown & Black Rust	1.125-1.5 kg	1.5-2.0 kg	750	-
Propiconazole 25% EC	Brown, Yellow & Black Rust	125	500	750	30
Tebuconazole 25% WG	Yellow Rust	0.1875	0.750	50	41
Zineb 75% WP	Rust	1.125-1.5 kg	1.5-2.0 kg	750-1000	-
Azoxystrobin 18.2% w/w+ Cyproconazole 7.3% w/w SC	Rust	0.26	1.0	500	50
Azoxystrobin 18.2% + Cyproconazole 7.3% w/w SC	Rust	0.26	1.0	500	50

Azoxystrobin 18.2% + Difenoconazole 11.4% w/w SC	Rust	0.03% or 0.3 g/l	0.1% or 1 ml / Litre water	500	35
Azoxystrobin 11% + Tebuconazole 18.3% w/w SC	Yellow Rust	82.5+137.25	750	500	-
Picoxystrobin 7.05% + Propiconazole 11.7% SC	Yellow Rust	200	1000	500	52
Pyraclostrobin 133 g/l + Epoxiconazole 50g/l SE	Yellow Rust	137.25	750	500	47
Tebuconazole 50% + Trifloxystrobin 25% WG	Yellow Rust	150+75	300	300-500	40
Triadimefon 25% WP	Rust	250	1.0	750	25
Imidacloprid 18.5% + Hexaconazole 1.5% FS	Rust	37:3	200	NA	Seed Dress or

Sources:

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